

PHYSICS-INFORMED TRAINING AT THE OPERATOR LEVEL

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ABSTRACT

We present a novel approach to train neural operators physics-informed, i.e., in a self-supervised fashion, purely relying on the governing physical equations. This enables training on infinite datasets where no sample is visited twice and removes the need to produce training data in an offline phase. While previous attempts for physics-informed training at the operator level were limited to benign problems and performed considerably worse than data-driven learning, our approach matches the gold-standard of data-driven training at a similar optimization cost. Our method is based on modifying the physics-informed loss functions to include a preconditioner such as a multigrid method to drastically reduce the ill-conditioning caused by unbounded PDE operators. We derive the improved conditioning theoretically, and illustrate the potential of the method for the Poisson, Allen-Cahn, and stationary Stokes equation, where it performs virtually identical to data-driven training at a comparable computational cost.

1 INTRODUCTION

Neural Operators Despite a century of development of numerical methods, partial differential equations (PDEs) remain expansive to solve, even more so in many-query situations, such as uncertainty quantification, optimal control, and inverse problems. The development of fast PDE solution techniques, e.g., through reduced-order/surrogate models therefore is of utmost importance. In the wake of the deep learning revolution, neural operators emerged as a powerful computational tool for surrogate modeling, drastically reducing PDE solution time in many applications. Neural operators approximate maps between infinite-dimensional spaces (operators) by a neural network, and are typically trained in a supervised fashion. They hence require a dataset that is built in an offline fashion, before they can be trained, and eventually used downstream. Data generation can constitute a substantial part of the computation and engineering effort, even more so in the context of physics foundation models that are pre-trained over an array of different PDE instances.

Physics-Informed Training Whenever the governing physics in the form of PDEs are known, neural operators can—in principle—be trained in a self-supervised fashion, relying on the physics residual or variational principles as a loss function. This approach is referred to as *physics-informed neural operator training*, and a natural idea present in the community since the inception of neural operators [1]. However, the success of physics-informed training of neural operators has remained limited to benign equations and learning tasks. More often than not, purely physics-informed learning fails to converge. In fact, supervised learning with its offline data-generation pipeline remains the gold standard for almost all applications. Nevertheless, physics-informed training offers unique advantages, such

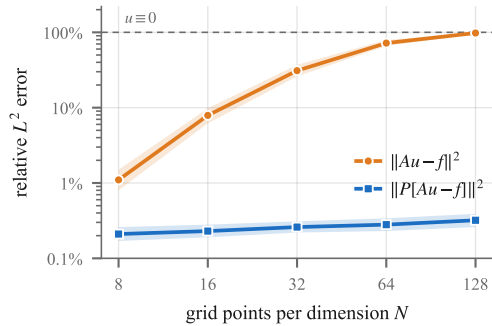


Figure 1: Adam on the plain versus the preconditioned Poisson least-squares loss, $N \times N$ grid. Placeholder numbers.

as the training in the infinite data-limit, where no sample is revisited, and, of course, it removes the need of an expensive dataset generation process.

Ill-Conditioning of PDE Operators In this manuscript, we show that the pathologies of physics-informed neural operator learning can be attributed to a single foundational property of PDE operators: PDE operators are unbounded maps, meaning they magnify their inputs without bounds, and thus lead to severe ill-conditioning in physics-informed loss functions. To make this concrete, take the derivative operator

$$\frac{d}{dx} : C^1(0, 1) \subset L^2(0, 1) \rightarrow L^2(0, 1), \quad u \mapsto u', \quad (1)$$

which amplifies oscillations: $x \mapsto \sin(\omega x)$ is bounded in $L^2(0, 1)$ independent of $\omega \in \mathbb{R}$, but its derivative $x \mapsto \omega \cos(\omega x)$ grows without bound in $L^2(0, 1)$ when $\omega \rightarrow \infty$. This phenomenon is inherent in all PDE operators. In the case of Poisson’s equation, and for a linear model that is the linear combination of a given basis set $\phi_1, \dots, \phi_N$, the physics-informed least-squares loss¹ on a single data point f takes the form

$$L_{\text{LS}}(u) = \frac{1}{2} \|Au - f\|^2, \quad A_{ij} = \int_{\Omega} \nabla \phi_j \nabla \phi_i \, dx, \quad (2)$$

where $A \in \mathbb{R}^{N \times N}$ is the stiffness matrix. The condition number of A scales like h^{-2} for standard bases, where h is the characteristic mesh-size. Conditioning thus deteriorates under mesh refinement, as higher oscillating frequencies can be resolved, the same manifestation of the unbounded nature of PDE operators stressed in 1. Note that $\nabla^2 L_{\text{LS}} = A^2$, hence the condition number governing the convergence speed of gradient descent (GD) scales like h^{-4} . Figure 1 illustrates that this ill-conditioning is so severe that training with first-order methods like GD and Adam becomes infeasible already for very coarse meshes, even in the benign case of a linear ansatz (no neural network at all). In the remainder of this article, we show that precisely this ill-conditioning renders physics-informed neural operator learning infeasible.

Preconditioned Physics-Informed Training To counter the ill-conditioning of the least-squares loss equation 2, we rely on *preconditioning*—precisely the tool the numerical analysis community uses to efficiently employ iterative methods in ill-conditioned settings in classical scientific computing. Abstractly, we *modify the least-squares loss* to include a preconditioner P

$$L_{\text{PLS}}(u) = \frac{1}{2} \|P[Au - f]\|^2, \quad (3)$$

where P maps from the right-hand side space to the solution space of the PDE. We stress that this construction is agnostic to the neural operator architecture employed and adds zero overhead to the inference once training finishes as it only influences the loss function.

On a conceptual level, the preconditioned loss equation 3 changes the norm in which the error is measured. For any given P we can recast the

$$L_{\text{PLS}}(u) = \frac{1}{2} \|P[Au - f]\|^2 = e^\top (A^\top P^\top P A) e$$

with the error $e = u - A^{-1}f$. This means a choice of P entails a norm in which e is measured, namely $A^\top P^\top P A$. The choice $P = A^{-1}$ recovers data-driven training and reduces the condition number of $\nabla^2 L_{\text{PLS}}$ to unity. In practice, is realized through fast approximate inverses $P \approx A^{-1}$, e.g., via multigrid methods []. Note that $P \approx A^{-1}$ is not the only choice; for saddle point problems, we alternatively employ block-diagonal preconditioners, Section ?? for details. Additionally, whenever applicable, we investigate energy formulations.

Main Contributions The main contributions are

- **Efficient physics-informed neural operator training.** We demonstrate that preconditioned physics-informed neural operator training allows data-free training of neural operators at comparable cost and accuracy of the gold standard of data-driven training. Our approach vastly outperforms previous approaches and is demonstrated for the Poisson, Allen-Cahn, and Stokes equation.

¹text

- **Flexibility and inference speed.** Our method can be combined with any neural operator architecture and once trained, adds no additional cost to the inference.
- **Training at infinite data limit.** Physics-informed training allows to practically approximate the infinite data limit, where no sample is visited twice. We demonstrate empirically that this is a favorable training paradigm for neural operators.

Related Literature Leave empty for now.

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The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing of 11 points. Times New Roman is the preferred typeface throughout. Paragraphs are separated by 1/2 line space, with no indentation.

Paper title is 17 point, in small caps and left-aligned. All pages should start at 1 inch (6 picas) from the top of the page.

Authors’ names are set in boldface, and each name is placed above its corresponding address. The lead author’s name is to be listed first, and the co-authors’ names are set to follow. Authors sharing the same address can be on the same line.

Please pay special attention to the instructions in section 4 regarding figures, tables, acknowledgments, and references.

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First level headings are in small caps, flush left and in point size 12. One line space before the first level heading and 1/2 line space after the first level heading.

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3.1.1 HEADINGS: THIRD LEVEL

Third level headings are in small caps, flush left and in point size 10. One line space before the third level heading and 1/2 line space after the third level heading.

4 CITATIONS, FIGURES, TABLES, REFERENCES

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The corresponding references are to be listed in alphabetical order of authors, in the REFERENCES section. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

4.2 FOOTNOTES

Indicate footnotes with a number² in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).³

4.3 FIGURES

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction; art work should not be hand-drawn. The figure number and caption always appear after the figure. Place one line space before the figure caption, and one line space after the figure. The figure caption is lower case (except for first word and proper nouns); figures are numbered consecutively.

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You may use color figures. However, it is best for the figure captions and the paper body to make sense if the paper is printed either in black/white or in color.

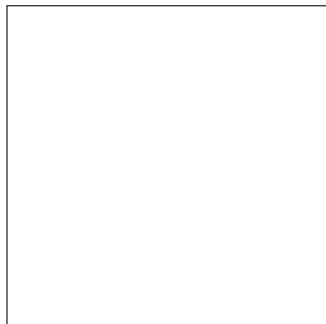


Figure 2: Sample figure caption.

²Sample of the first footnote

³Sample of the second footnote

Table 1: Sample table title

PART	DESCRIPTION
Dendrite	Input terminal
Axon	Output terminal
Soma	Cell body (contains cell nucleus)

4.4 TABLES

All tables must be centered, neat, clean and legible. Do not use hand-drawn tables. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

5 DEFAULT NOTATION

In an attempt to encourage standardized notation, we have included the notation file from the textbook, *Deep Learning* Goodfellow et al. (2016) available at https://github.com/goodfeli/dlbook_notation/. Use of this style is not required and can be disabled by commenting out `math_commands.tex`.

Numbers and Arrays

a	A scalar (integer or real)
$\mathbf{a}$	A vector
$\mathbf{A}$	A matrix
$\mathbf{A}$	A tensor
$\mathbf{I}_n$	Identity matrix with n rows and n columns
$\mathbf{I}$	Identity matrix with dimensionality implied by context
$\mathbf{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
$\text{diag}(\mathbf{a})$	A square, diagonal matrix with diagonal entries given by $\mathbf{a}$
a	A scalar random variable
$\mathbf{a}$	A vector-valued random variable
$\mathbf{A}$	A matrix-valued random variable

Sets and Graphs

270	$\mathbb{A}$	A set
271	$\mathbb{R}$	The set of real numbers
272	$\{0, 1\}$	The set containing 0 and 1
273	$\{0, 1, \dots, n\}$	The set of all integers between 0 and n
274	$[a, b]$	The real interval including a and b
275	$(a, b]$	The real interval excluding a but including b
276	$\mathbb{A} \setminus \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
277	$\mathcal{G}$	A graph
278	$Pa_{\mathcal{G}}(\mathbf{x}_i)$	The parents of $\mathbf{x}_i$ in $\mathcal{G}$

Indexing

285	a_i	Element i of vector $\mathbf{a}$, with indexing starting at 1
286	$\mathbf{a}_{-i}$	All elements of vector $\mathbf{a}$ except for element i
287	$A_{i,j}$	Element i, j of matrix $\mathbf{A}$
288	$\mathbf{A}_{i,:}$	Row i of matrix $\mathbf{A}$
289	$\mathbf{A}_{:,i}$	Column i of matrix $\mathbf{A}$
290	$\mathbf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor $\mathbf{A}$
291	$\mathbf{A}_{:,:,i}$	2-D slice of a 3-D tensor
292	$\mathbf{a}_i$	Element i of the random vector $\mathbf{a}$

Calculus

295	$\frac{dy}{dx}$	Derivative of y with respect to x
296	$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
297	$\nabla_{\mathbf{x}} y$	Gradient of y with respect to $\mathbf{x}$
298	$\nabla_{\mathbf{X}} y$	Matrix derivatives of y with respect to $\mathbf{X}$
299	$\nabla_{\mathbf{x}} y$	Tensor containing derivatives of y with respect to $\mathbf{X}$
300	$\frac{\partial f}{\partial \mathbf{x}}$	Jacobian matrix $\mathbf{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
301	$\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ or $\mathbf{H}(f)(\mathbf{x})$	The Hessian matrix of f at input point $\mathbf{x}$
302	$\int f(\mathbf{x}) d\mathbf{x}$	Definite integral over the entire domain of $\mathbf{x}$
303	$\int_{\mathbb{S}} f(\mathbf{x}) d\mathbf{x}$	Definite integral with respect to $\mathbf{x}$ over the set $\mathbb{S}$

Probability and Information Theory

324	$P(a)$	A probability distribution over a discrete variable
325	$p(a)$	A probability distribution over a continuous variable, or
326		over a variable whose type has not been specified
327		
328	$a \sim P$	Random variable a has distribution P
329	$\mathbb{E}_{x \sim P}[f(x)]$ or $\mathbb{E}f(x)$	Expectation of $f(x)$ with respect to $P(x)$
330	$\text{Var}(f(x))$	Variance of $f(x)$ under $P(x)$
331	$\text{Cov}(f(x), g(x))$	Covariance of $f(x)$ and $g(x)$ under $P(x)$
332	$H(x)$	Shannon entropy of the random variable x
333	$D_{\text{KL}}(P Q)$	Kullback-Leibler divergence of P and Q
334	$\mathcal{N}(x; \mu, \Sigma)$	Gaussian distribution over x with mean μ and covariance
335		Σ
336		
337		
338		

Functions

340	$f : \mathbb{A} \rightarrow \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$
341	$f \circ g$	Composition of the functions f and g
342	$f(x; \theta)$	A function of x parametrized by θ . (Sometimes we write
343		$f(x)$ and omit the argument θ to lighten notation)
344		
345	$\log x$	Natural logarithm of x
346	$\sigma(x)$	Logistic sigmoid, $\frac{1}{1 + \exp(-x)}$
347	$\zeta(x)$	Softplus, $\log(1 + \exp(x))$
348	$\ x\ _p$	L^p norm of x
349	$\ x\ $	L^2 norm of x
350	x^+	Positive part of x , i.e., $\max(0, x)$
351	$\mathbf{1}_{\text{condition}}$	is 1 if the condition is true, 0 otherwise
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6 FINAL INSTRUCTIONS

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the REFERENCES section; see below). Please note that pages should be numbered.

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Otherwise, please generate your PostScript and PDF files with the following commands:

```
dvips mypaper.dvi -t letter -Ppdf -G0 -o mypaper.ps
ps2pdf mypaper.ps mypaper.pdf
```

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Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package.

Always specify the figure width as a multiple of the line width as in the example below using .eps graphics

```
\usepackage[dvips]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.eps}
```

or

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

for .pdf graphics. See section 4.4 in the graphics bundle documentation (<http://www.ctan.org/tex-archive/macros/latex/required/graphics/grfguide.ps>)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the \- command.

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AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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A APPENDIX

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